

Analytic Calculation of an Axial-Flux Permanent Magnet Motor Torque

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Abstract: This paper proves that fast semi-analytic methods can be applied to perform the three dimensional design calculation of axial-flux permanent magnet machines successfully. The analytic calculation can be summarized into the following three steps: Firstly, each axial-flux permanent-magnet machine pole pair – or several pole pairs in the case of fractional slot windings – is divided on the air gap surface radially into sectors-of-rings, which then form the air gap surfaces of elementary machines. Secondly, the behaviour of each elementary machine is calculated individually assuming a two-dimensional flux problem. Thirdly, the machine properties are found by summing the results given by all the elementary machines and all pole pairs of the machine.

1 Introduction

Speed controlled low speed direct drive systems are sensitive to deficiencies in the torque quality. The motor torque vibration amplitudes should, according to the experience of the authors at frequencies less than 100 Hz, be no more than 0.5 % of the rated torque to avoid harmful noise and e.g. uncomfortable ride in a direct-driven elevator.

Traditionally, motors have been designed using analytic calculations and experimental knowledge. Analytic design equations are nowadays calculated by the help of computers in everyday design. Also the Finite Element Analysis (FEA) has become a common tool.

Even with powerful present-day computer technology accurate, three-dimensional (3D) modelling of the motors remains a time consuming process. With an analytic model of the motor the calculation results obtained may be less accurate than those obtained with FEA, but the results required can be reached fast.

Even though an axial flux machine in many cases is not an electro-magnetically ideal solution it offers a very attractive solution in several cases, especially, when an electrical machine has to be directly integrated to a mechanical machine. Often an axial-flux motor requires less material than a radial flux motor in the same application. Sitapati and Krishnan [1] compared radial and axial flux topologies and found that the axial flux motor has a higher power density compared to radial flux motors. Zhang & al. [2] came to the same conclusion concerning the torque density. Also Parviainen & al. [3] have reached similar results. The behaviour of PM-motors has been studied widely. Different methods for motor analysis and torque ripple reduction have been developed and studied by many authors. E.g. Ishikawa [9], Jahns [10], Studer [11], Zhu [12, 13] and Bianchi & Bolognani [14] have been active in the field.

2 The calculation model

The calculation model used here is based on dividing the axial-flux machine into elementary machines, which physically are radial sectors of rings, and assuming two-dimensional fields in each individual elementary machine. The two-dimensionality assumption may be criticized because of the curvy nature of the elementary machines and because of the flux fringing at the inner and outer radii of the machine. According to the experience of the authors it, however, does not cause a remarkable error. The calculation commences at the inner radius $r_{in} = r_0$ of the machine and enters step vice to the outer radius $r_{out} = r_{ni}$ of the machine. The motor geometry and the materials are given by the designer and the model is fed with three phase currents, which may have an arbitrary shape and amplitude. In Fig. 1 is illustrated the division of a sinusoidally formed permanent magnet into n_i sectors of rings each of which belong to one elementary machine.

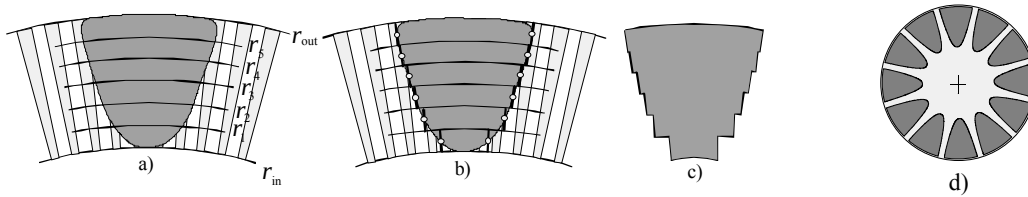


Figure 1. Division of a magnet with sinusoidal contour for six elementary machines $n_i = 6$. a) original permanent magnet form, b) sectors of rings c) widths of elementary magnets of the elementary machines and d) an axial flux rotor equipped with sinusoidal magnets.

The coarse division obtained by using a small number ($n_i = 6$) of radial elementary machines (Fig. 1c) differs remarkably from the results obtained with the original shape (Fig. 1 a) and the harmonic

torque calculation may give erroneous results. A coarse division of the magnets may produce unexpected torque behaviour since each magnet sector of a ring has edges parallel to the stator slots. Each elementary machine has often also an individual permanent magnet width $w_{PM,ni}$ which also may result in different relative permanent magnet widths $\alpha_{PM,ni}$ compared to the pole pitch $\tau_{p,ni}$. The width $w_{PM,ni}$ is defined at the average radius $(r_i + r_{i+1})/2$ of the elementary machine

$$\alpha_{PM,ni} = \frac{w_{PM,ni}}{\tau_{p,ni}} \pi. \quad (1)$$

The pole pitch $\tau_{p,ni}$ of an elementary machine depends on the amount of pole pairs p and equals

$$\tau_{p,ni} = \frac{\pi(r_i + r_{i+1})/2}{p}. \quad (2)$$

2.1 Air-Gap Flux Created by the Permanent Magnets

The form and the amplitude of the permanent magnet created flux density in the air-gap depends on the geometry and the materials of the magnetic circuit, the length of the air-gap and the magnetomotive force of the magnet.

Chung and Gweon [4] introduced a Fourier-analysis-based equation for an air-gap flux density distribution created by the permanent magnets. It is adopted here in the following modified form

$$B_{PM,ni} = \sum_{n=1,3,5,\dots}^{\infty} \frac{\frac{8B_r}{n\pi} \sin\left(\frac{n\pi\alpha_{PM,ni}}{2}\right)}{\left(e^{\frac{-2n\pi\delta}{\tau_{p,ni}}} + 1 \right) + \frac{\mu_m \left(-e^{\frac{-2n\pi\delta}{\tau_{p,ni}}} + 1 \right) \left(e^{\frac{2n\pi h_{PM}}{\tau_{p,ni}}} + 1 \right)}{\mu_0 \left(e^{\frac{2n\pi h_{PM}}{\tau_{p,ni}}} - 1 \right)}} e^{\frac{-n\pi\delta}{\tau_{p,ni}}} \cos\left(\frac{n\pi x}{\tau_{p,ni}}\right). \quad (3)$$

B_r is the permanent magnet remanence flux density, h_{PM} the permanent magnet thickness, $\alpha_{PM,ni}$ relative permanent magnet width compared to the pole pitch $\tau_{p,ni}$ and δ is the air gap length. Eq. (3) ignores the effects of slotting on the flux density distribution. The stator slotting effect is traditionally taken into account by just using the Carter factor. In this case, however, it is not sufficient to use the Carter factor but a permeance function of the air gap region is required. Heller and Hamata [5] present different equations for the air gap permeance function k_Λ , depending on the air-gap length, slot and

slot opening parameters. The following equation introduced by Weber [6] is one of the most appropriate equations referred to by Heller and Hamata

$$B(\alpha) = k_{\Lambda} B_{\max} = \left(1 - 2\beta \sin^{2n} \frac{\pi}{\alpha_d} \alpha \right) B_{\max}, \quad n = \frac{\tau_s - w_{so}}{w_{so}}$$

$$\beta = (B_{\max} - B_{\min}) / 2B_{\max} = \frac{1 + u^2 - 2u}{2(1 + u^2)}, \quad u = \frac{w_{so}}{2\delta} + \sqrt{1 + \left(\frac{w_{so}}{2\delta} \right)^2}. \quad (4)$$

α_d is the angle ($0 \dots \pi$) corresponding to slot pitch τ_s . The depth of the drop in the flux density is ($B_{\max} - B_{\min}$). $\alpha \in [0, \pi]$ under a slot pitch τ_s .

To compare the applicability of Eqs. (3) And (4) an example was calculated and compared to a 3D FEA of the same model (Fig. 2).

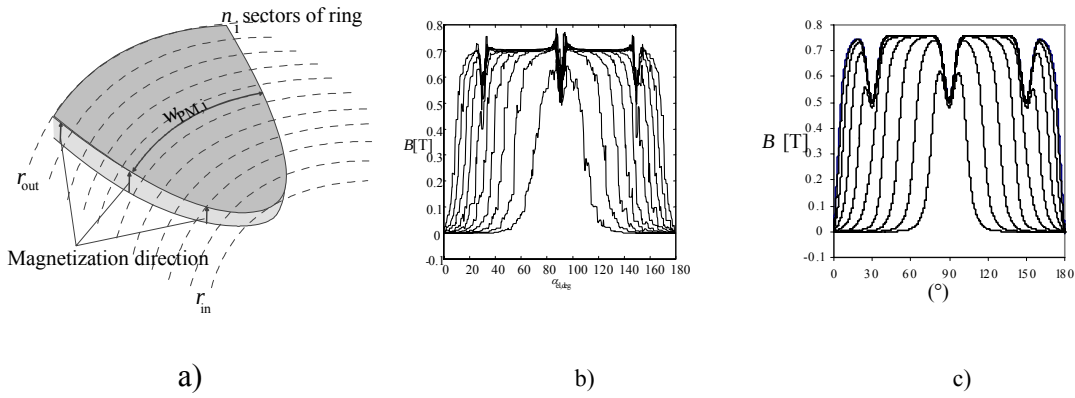


Figure 2. a) Sinusoidal permanent magnet (3D), b) corresponding air-gap flux density curves at stator surfaces of ten elementary machines ($n_i = 10$) in an axial flux motor found by FEA and c) air-gap flux density calculated using Eqs. (3) and (4) on the stator surface. Please notice the effects of three slot openings in the distribution. 180 degrees corresponds to one pole pitch.

The results calculated with eqns. (3) and (4) correspond well to the results found by FEA, especially the flux integrated over a pole gives good correspondence. The flux density spikes at slot edges are, however, not found by the analytic calculation. The amount of flux in these spikes is, however, negligible to the performance of the motor.

The analytic calculation is simple and fast if the shape of the magnet is a sector of a ring (SOR) having the same relative magnet width α_{PM} at every radial position thus having no skew related to the stator slots. When the shape of the magnets is more complicated – such as the sinusoidal magnet in Fig. 2 – the calculation has to be done in several sectors of a ring (elementary machines).

2.2 Magnetic Circuit

The magnetic circuit of an elementary motor is modelled using a reluctance network, which takes the saturation of the iron into account. Fig. 3 shows the reluctance network of an elementary machine having six slots per pole, $q = 2$ (slots per pole and phase).

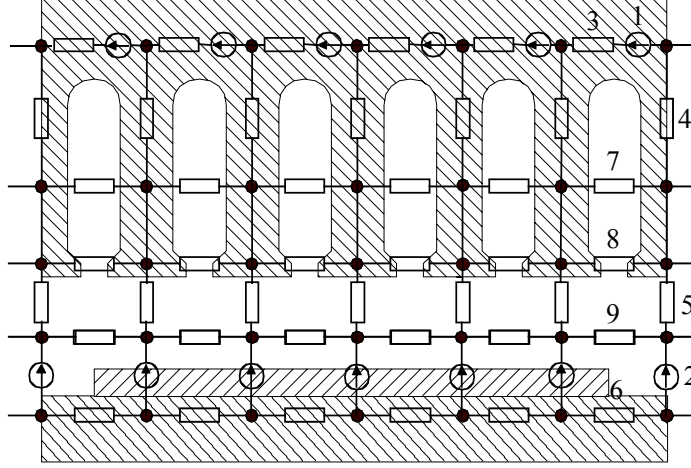


Figure 3. Two-dimensional reluctance network of one pole area of an elementary machine. The components are: 1 mmf of the winding in the slot, 2 mmf of the rotor (permanent magnet), the reluctance of the 3. stator yoke, 4. stator tooth, 5. air-gap, 6. rotor yoke, 7. stator slot, 8. stator slot opening and 9. tangential leakage reluctance in the air-gap

The reluctance network is, in this case, the simplest possible to model the magnetic circuit. Adding the nodes and reluctance components increases the reluctance matrix rapidly. The results given later prove this simple network to give accurate enough results compared to FEA results.

In case of an integer slot winding the elementary machines form one pole pair of the machine. In the case of a fractional slot winding or in cases in which non-uniformity in the magnet positions or magnet sizes are included several pole pairs must be calculated.

The main route of the flux goes along the stator yoke, the teeth, the air-gap and the rotor yoke. The BH -curve of the magnetic material is defined, and the values of the reluctances are calculated depending on the saturation of the magnetic circuit components. The leakage reluctances are kept as constants.

The flux φ in each circuit is solved by

$$[\varphi] = [R]^{-1} [Ni], \quad (5)$$

where $[R]$ is the reluctance matrix corresponding to the reluctance network given in Fig. 3 and $[Ni]$ is the mmf matrix.

2.3 Torque Calculation

There are different possibilities of calculating the torque. Knowing the instantaneous local flux densities $\vec{B}(t)$ and instantaneous currents $i(t)$ (direction $d\vec{l}$) makes the Lorentz force well applicable

$$d\vec{F}(t) = di(t)(d\vec{l} \times \vec{B}(t)). \quad (6)$$

The Maxwell stress tensor may be calculated only in FEA. The stress tensor method easily fails in the torque calculation. In our case the accuracy of the air gap flux density components is far from accurate enough for the calculation of Maxwell stresses accurately. The Lorentz force based methods seem to give satisfactory results.

Air Gap Flux Linkage

The instantaneous power $P(t)$ may be together with air gap flux linkage ψ_m be used in the torque calculation

$$P(t) = \omega T_e(t) = u(t)i(t) \Leftrightarrow T_e(t) = \frac{u(t)}{\omega} i(t) = \psi_m(t)i(t) \quad (7)$$

In order to take the motor space harmonics into account in the torque production the instantaneous component values may be used and thus the torque of an elementary machine may be expressed as a sum of the torques of all individual coils as:

$$T_{e, \text{elem}, ni}(t) = \sum_{ni=1}^{n_c} \xi N \varphi_{ni}(t) i_{ni}(t). \quad (8)$$

$\xi N \varphi_{ni}(t) = \psi_{ni}(t)$ is the flux linkage component of a coil of the elementary machine having ξN effective turns, $i_{ni}(t)$ is the instantaneous current of an individual coil of the elementary machine and n_c is the number of coils. The torque $T_{e,p}(t)$ of the whole pole pair - or in the case of fractional slot winding the whole symmetric sector - is found by summing the torques of the elementary machines

$$T_{e,p}(t) = \sum_{ni=1}^{n_i} T_{e, \text{elem}, ni}(t) \quad (9)$$

and multiplying the result by the amount of similar sectors of the machine.

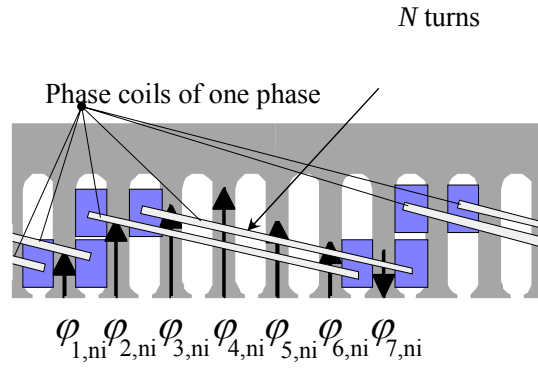


Figure 4. The flux components $\varphi_{n,ni}$ and the phase coils of one phase in a 5/6-pitched short pitch winding, where $q = 2$. The fluxes $\varphi_{2,ni} \dots \varphi_{6,ni}$ pass through one of the coils having N turns and the fluxes $\varphi_{3,ni} \dots \varphi_{7,ni}$ through another coil with N turns forming the air gap flux linkage ψ_m components.

In the reluctance network model the flux $\varphi_{i,ni}$ in each part of the magnetic circuit is known. This means that also the flux linkage $\psi_{i,ni}(t)$ in each of the coils as well as the currents in each of the coils is known during every time step. Without making a big mistake $\xi \approx 1$ may be assumed.

In practice, part of a flux $\varphi_{i,ni}(t)$ goes through the coil window, producing the flux linkage only with part of the turns of the coil $\xi < 1$. In a case where high accuracy is needed the slot area should be divided in several reluctance components.

Cogging Torque

Because of the permeance variation in the slot opening the flux enters the teeth creating tangential forces in the rotor (Fig. 5). The slot opening geometry is the most dominating factor in cogging.

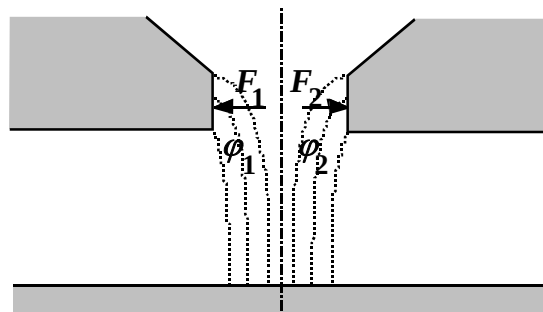


Figure 5. The air-gap flux components under a slot opening that may create cogging.

The flux components on different sides of the opening are often not - depending on the permanent magnet position - equal and thus the forces F_1 and F_2 do not compensate each other. We assume that each slot opening may be divided in two equal halves carrying individual amounts of flux φ_1 and φ_2 (Fig. 5) entering the teeth tangentially on the slot opening sides. These flux components are calculated

using the air-gap flux density distribution under the slot opening. Each force component $F_j(t)$ can be calculated as

$$F_j(t) = \frac{B_j^2(t)A}{2\mu_0} = \frac{\phi_j^2(t)}{2A\mu_0}; \quad A = \frac{w_{so} L_{sc}}{2n_i} \quad (10)$$

where w_{so} is the width of the side of the slot opening, L_{sc} the length of the stator stack and n_i the number of elementary machines. The forces of each slot produce a single cogging torque component

$$T_{\text{cog},ni}(t) = (F_1(t) - F_2(t)) \frac{r_i + r_{i+1}}{2} \quad (11)$$

The total cogging torque is caused by all the elementary machines n_i in all stator slots Q_s

$$T_{\text{cog}}(t) = \sum_{Q_s} \sum_{ni} T_{\text{cog},ni}(t) \quad (12)$$

3 Comparison between FEA and Analytic Calculation

The simplest shape of a permanent magnet that can be used in axial flux motors is a sector of a ring having radial length of $r_{\text{out}} - r_{\text{in}}$. The cogging torque with SOR (relative width $0.9\tau_p$) magnets was calculated using both 3D Finite-Element-Method (FEM) based program (Flux3D™) and the analytic method given above, Fig. 6.

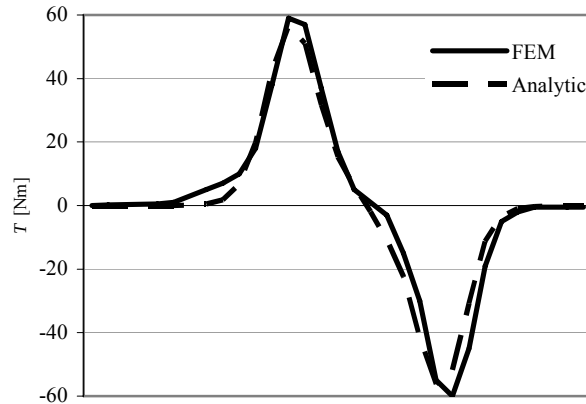


Figure 6. Calculated cogging torque along one slot pitch created by SOR-magnets (relative width $0.9\tau_p$).

The shapes and amplitudes of cogging seem to be close to each other. The simple semi-analytic method described above seems to be valid for the calculation of the instantaneous torque of an axial flux permanent magnet machine. The 3D FEM calculation model used is illustrated in Fig. 7. Fig. 8 shows the torque behaviour along one pole pitch and Fig. 9 the frequency contents of the torque. The

calculation was performed for a motor having the following parameters:

Table 1 Calculation parameters

Motor parameter	
power (P_n)	8.0 kW
rated voltage (U_n)	400 V
rated current (I_n)	17 A
supply frequency (f_s)	18 Hz
Number of stator slots (Q_s)	120
Number of pole pairs (p)	10
Permanent magnets:	
- Material	NdFeB
- Remanence flux density	1.15 T
- Relative permeability	1.2
Stator dimensions:	
- Outer diameter	600 mm
- Inner diameter	450 mm

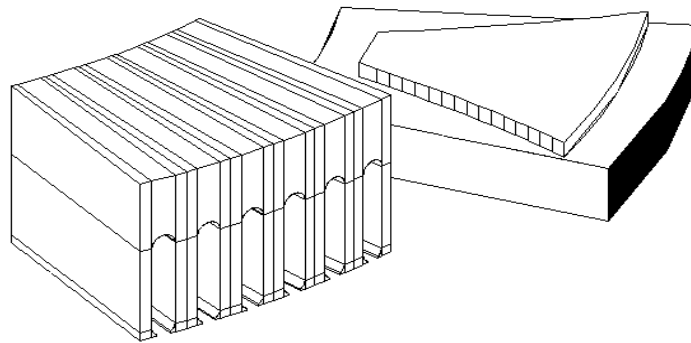


Figure 7. 3D finite element model of the AFPMSM used in the reference calculations.

The finite element calculation was done using a time step corresponding to 122 points per one pole pair length. In order to have equal conditions for both calculation processes the analytic calculation was done correspondingly.

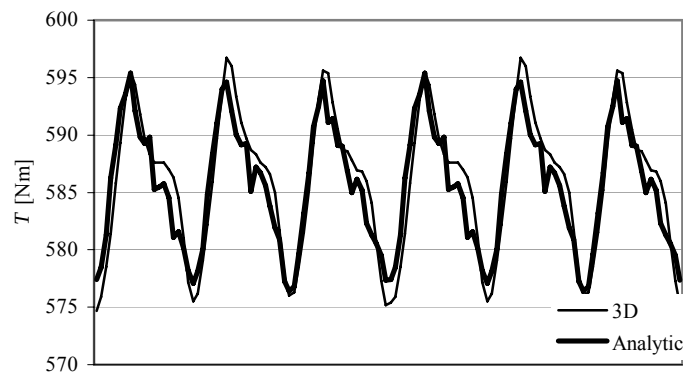


Figure 8. The calculated torque for the reference motor along one pole pair length. The thin line is calculated with 3D FEM and the bold line with the analytic equations.

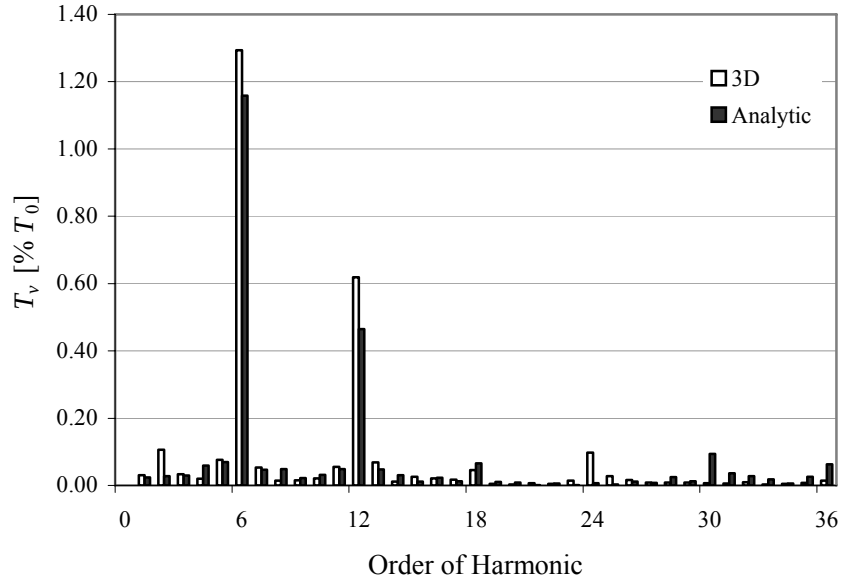


Figure 9. Harmonic contents of the torque curves (Fig. 8) calculated with 3D FEM and with the analytic model.

The shapes of the torque and especially the harmonic contents are very similar. The shape of the air-gap flux density waves at different radii of the motor (producing the torques in Fig. 8) are given in Fig. 10.

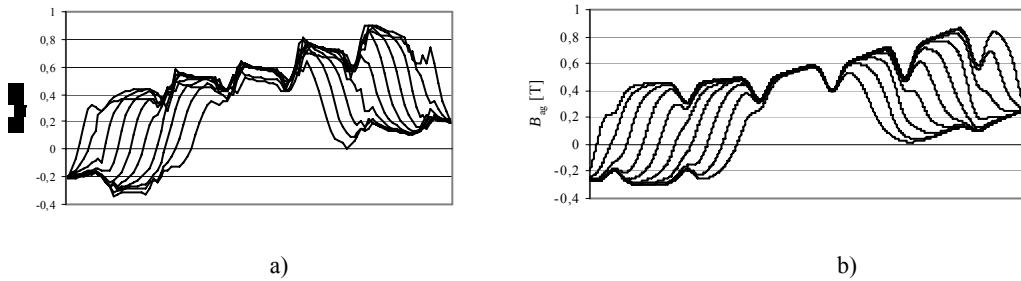


Figure 10. Air-gap flux density B_{ag} shapes in one pole area in ten elementary machines at different radii in the AFPMSM, corresponding to the load condition of Fig. 10, a) calculated with 3D FEM, b) calculated with analytic equations.

It can be observed some non-uniformity between the flux density distributions. Although the two flux density forms above are slightly different, it is shown in Figs. 8 and 9 that the torques produced in both cases are quite equivalent.

In addition to the results, also the hardware requirements and required time for the model definition and the problem calculation are essential. To verify the differences between the two methods the main parameters of the reference calculation and system requirements are listed in table 2:

Table 2 Model parameters and system requirements

Parameter	Flux 3D Software by CEDRAT	Analytic calculation
Model size	One pole	One pole pair
Number of 3D nodes	290 300	
Number of 3D elements	213 110	
Elements	2 nd order (Hierarchic) *)	
Air-gap element size	1.25 mm (two layers)	
Iteration criteria		
- Newton-Rhapson (non-lin.)	10 ⁻⁴	
- Stabilisation (linear system)	10 ⁻⁶	8 loops in saturation
Data storage size:		
- Geometry	393 kbytes	< 1 kbyte
- Element network	111 Mbytes	
- Results	8.6 Mbytes / calculation point	< 1 kbyte
Computer requirements:		
- Processor recommendation **)	Min. 1 GHz	100 MHz
- Random Access Memory **)	1 Gbyte	126 Mbytes
Calculation time (Pentium 4, 1.8 GHz / 1 Gbyte) :		
- pre-calculation	~ 58 min	< 10 s
- calculation / point	~ 40 min	< 1.5 s

*) Cubic and pyramid elements

**) values are not definite minimum requirements, but recommended for reasonable calculation time.

The analytic method presented in this work is conclusively faster both in the problem definition and in the calculation. Due to the simplicity of the reluctance network the accuracy is not as good as with 3D FEM.

4 Conclusions

The results have proven that the analytic method described here can be used for axial-flux PM-machine modelling. It cannot compete in accuracy with 3D FEA, but for every-day design calculations it may be applied and, especially, as a pre study for FEA it is valid.

Even though the structure of the reluctance network used appears to be simple the calculation results are promising when comparing them with the FEA results and with measurements of actual machines. This may be mainly explained by the fact that even a simple network is capable of calculating reliably enough the instantaneous flux in the teeth and thus the winding harmonics and the torque.

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